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APOGEE

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APOGEE performs a systematic survey of all Milky Way stellar components using high-resolution, multi-object, near-infrared spectroscopy.

The APOGEE Survey

APOGEE (<https://ui.adsabs.harvard.edu/abs/2017AJ....154...94M/abstract>) is a large-scale, stellar spectroscopic survey conducted in the near-infrared (H-band (/instruments/apogee_spectrograph/#TheAPOGEENorthernandSouthernSpectrographs)) portion of the electromagnetic spectrum. APOGEE consists of two 300-fiber cryogenic spectrographs (/instruments/apogee_spectrograph/) that operate onboard the 2.5-meter Sloan Foundation Telescope (</instruments/#Telescopes>) and the 1-meter NMSU Telescope (</instruments/#Telescopes>) at Apache Point Observatory (APO) in New Mexico, United States and the 2.5-meter Irénée du Pont Telescope (</instruments/#Telescopes>) of Las Campanas Observatory (LCO) in Atacama de Chile. APOGEE is unique among large spectroscopic surveys because it is able to acquire high-resolution ($R \sim 22,500$) spectra across the entire Milky Way.

These pages provide a brief introduction to the APOGEE survey and its data products. See the APOGEE technical papers (<https://www.sdss4.org/dr17/irspec/technical-papers-and-additional-references/>) for detailed information on all aspects of the APOGEE survey and advice on how to cite these in your work.

The current data release is Data Release 17 (DR17) and this presents the last new observations from the APOGEE-2 project with homogenous processing of all APOGEE-1 and APOGEE-2 data. Users already familiar with APOGEE data can visit the DR17 Synopsis page (https://www.sdss4.org/dr17/irspec/dr_synopsis) for a summary of how the data analysis and data products have changed.

APOGEE Data Products

There are 5 major data products (/dr17/irspec/spectro_data/) provided by the APOGEE survey:

- ☆ **Spectra** - Three different types of reduced spectra (/dr17/irspec/spectro_data/#SpectralData) are available: visit spectra, combined spectra, and ASPCAP spectra. Descriptions for these three types of spectra and how to use them can be found on [Using Spectra \(/dr17/irspec/spectra/\)](/dr17/irspec/spectra/).
- ☆ **Radial Velocities (RVs)** - RVs can be found in the spectra files themselves, as well as in the allVisit (/dr17/irspec/spectro_data/#SummaryCatalogs) catalog (data model (https://data.sdss.org/datamodel/files/APOGEE_ASPCAP/APRED_VERS/ASPCAP_VERS/allVisit.html)) and allStar (/dr17/irspec/spectro_data/#SummaryCatalogs) catalog (data model (https://data.sdss.org/datamodel/files/APOGEE_ASPCAP/APRED_VERS/ASPCAP_VERS/allStar.html)) for the individual visit RVs and combined RVs, respectively. For an introduction on how to use RVs, go [Using Radial Velocities \(/dr17/irspec/use-radial-velocities/\)](/dr17/irspec/use-radial-velocities/).
- ☆ **Stellar Parameters** - Stellar parameters, including effective temperature, surface gravity, metallicity are derived using the APOGEE Stellar Parameters and Chemical Abundance Pipeline (ASPCAP (</dr17/irspec/aspcap/>)). Stellar Parameters can be found in the aspcapStar spectra files as well as the allStar summary file (/dr17/irspec/spectro_data/#SummaryCatalogs) (data model (https://data.sdss.org/datamodel/files/APOGEE_ASPCAP/APRED_VERS/ASPCAP_VERS/allStar.html)). For instructions on how to use APOGEE stellar parameters, review [Using Stellar Parameters \(/dr17/irspec/parameters/\)](/dr17/irspec/parameters/).
- ☆ **Chemical Abundances** - ASPCAP also provides chemical abundances for 18 elements. However, the precision and accuracy of these abundances vary based on stellar parameters. Chemical Abundances can be found in the allStar summary file (/dr17/irspec/spectro_data/#SummaryCatalogs) (data model (https://data.sdss.org/datamodel/files/APOGEE_ASPCAP/APRED_VERS/ASPCAP_VERS/allStar.html)) in the "{X}_Fe" tags. Users should consult the [Using Chemical Abundances \(/dr17/irspec/abundances/\)](/dr17/irspec/abundances/) page for how to properly use APOGEE chemical abundances.
- ☆ **Samples** - APOGEE has carefully constructed its stellar samples to enable a user to translate between observations and underlying populations using its selection function. An overview of this information can be found in [Using Targets/Samples \(/dr17/irspec/targetingbits/\)](/dr17/irspec/targetingbits/).

The RVs, stellar parameters, and chemical abundances can all be found in the allStar summary file

(/dr17/irspec/spectro_data/#SummaryCatalogs) (data model

(https://data.sdss.org/datamodel/files/APOGEE_ASPCAP/APRED_VERS/ASPCAP_VERS/allStar.html)). Be sure to read and examine the [Using](#) pages for instructions from the APOGEE team on how these quantities should be used and analyzed.

For instructions on how to get data from the SAS, SAW, or CAS, see the data access (/dr17/irspec/spectro_data/) page. For some examples of working with the data via the the SAS (FITS files accessed via a programming language), SAW (GUI web app), and/or CAS (SQL), see the examples (</dr17/irspec/catalogs/>) page.

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APOGEE observes a few hundred stars towards the Galactic bulge with the 2.5-meter Sloan Telescope during the full moon. The 1-meter NMSU telescope is in the foreground, with the lights of El Paso on the horizon.

Image Credit: S. R. Majewski

Observing and Data Use Information

APOGEE Observing (</dr17/irspec/apogee-observing/>)

gives an overview of how APOGEE spectroscopic data are taken.

APOGEE Data Access (/dr17/irspec/spectro_data/)

describes all of the APOGEE data products that are available through the Science Archive Server.

Examples (</dr17/irspec/catalogs/>)

provides examples of how to query the database or summary data files in order to retrieve a variety of information for different types of targets.

Using APOGEE Stellar Parameters (</dr17/irspec/parameters/>)

describes things you need to know if you plan to use the derived stellar atmospheric parameters.

Using APOGEE Chemical Abundances (</dr17/irspec/abundances/>)

describes things you need to know if you plan to use the derived elemental abundances.

Using APOGEE Spectra (</dr17/irspec/spectra/>)

describes features in APOGEE spectra about which anyone looking at spectra should be aware.

Using APOGEE Radial Velocities (</dr17/irspec/use-radial-velocities/>)

describes things you need to know if you plan to use the APOGEE radial velocities.

Using APOGEE Targets/Samples (</dr17/irspec/targettingbits/>)

describes things you need to know if you plan to use the APOGEE targeting information.

Software and Processing Information

APOGEE Target Information (/dr17/irspec/targets/)

describes the way in which APOGEE targets are chosen and how this is documented in the target flags.

APOGEE Visit Spectra Reduction (/dr17/irspec/apred/)

gives information about how individual visit spectra are observed, processed, and stored.

APOGEE Visit Spectra Combination (/dr17/irspec/spectral_combination/)

gives information about how the combined spectra for each star are created and stored.

APOGEE Radial Velocities (/dr17/irspec/radialvelocities/)

describes the derivation of radial velocities from both individual visit and final combined spectra.

APOGEE Stellar Parameter and Abundance Determination (/dr17/irspec/aspcap/)

describes how the stellar parameters and abundances available in APOGEE spectroscopic catalogs are derived.

APOGEE Spectral Libraries (/dr17/irspec/apogee-libraries/)

describes the spectral grids used in ASPCAP.

APOGEE Caveats (/dr17/irspec/caveats/)

gives a running list of known issues with this release.

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Additional References

- ☆ APOGEE Bitmasks (/dr17/irspec/apogee-bitmasks/) are fully described on the APOGEE Bitmasks page and also are given with the full listing of SDSS Bitmasks (/dr17/algorithms/bitmasks/) alongside a primer on their use.
- ☆ Glossary (/dr17/help/glossary/) describes common terms used with SDSS operations and data.
- ☆ Technical papers and references (/dr17/irspec/technical-papers-and-additional-references/) describe how to cite survey publications in your papers, lists all technical papers, and also provides references for the catalogs used in targeting.

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A Brief History of APOGEE

- ☆ As part of the planning for “After Sloan II” (AS2), it was felt that more mileage could come from the Sloan infrastructure investment by extending survey operations into bright time. An invitation to submit ideas for potential bright time experiments was extended to Michael Skrutskie (University of Virginia) by the AS2 Steering Committee. In August 2005, Skrutskie and John Wilson (Univ. of Virginia) submitted a white paper (Skrutskie & Wilson 2015 (<https://ui.adsabs.harvard.edu/abs/2015arXiv150308918S/abstract>)) describing the potential of high throughput, multifiber, NIR spectroscopy on the Sloan 2.5-m telescope to the Astrophysical Research Corporation (ARC) Futures Committee.
- ☆ Around the same time, Steven Majewski (also University of Virginia) was attending an Aspen Center workshop, where Katia Cunha (Steward Observatory and Observatorio Nacional) presented a multi-element analysis of Galactic radial metallicity gradients (from Cunha & Daflon 2004 (<https://ui.adsabs.harvard.edu/abs/2004ApJ...617.1115D/abstract>)). This work combined nebular abundances of H II regions and photospheric abundances of young OB stars. These two tracers are bright and, thus, are accessible to spectroscopy at optical wavelengths, despite the heavy dust obscuration at low Galactic latitudes; these tracers are, however, severely limited in number and biased towards young populations. Though state of the art at the time, this study was based on only several dozen sources inhomogeneously sampling the disk from 5-13 kpc from the Galactic center, and highly biased to very young sources. Moreover, the results of the study did not match the chemical evolution models of the Milky Way.
- ☆ At the same time, Majewski was conducting a large, one-star-at-a-time optical, low-resolution spectroscopic survey (the Grid Giant Star Survey). This survey identified bright but distant giant stars across the sky to serve as the astrometric grid for the (soon defunct) NASA Space Interferometry Mission. Building from that and inspired by Cunha’s presentation, Majewski suggested that an industrial-scale survey of the Milky Way with a specific focus on the previously much-neglected low-latitude Milky Way was timely, to address questions like those presented by Cunha in a comprehensive and statistically robust way. This aim necessitated working in the infrared to break through the thick veil of interstellar dust, moving to high resolution for accurate chemical abundance determinations, and focusing on

giant stars, which are intrinsically bright and therefore accessible to large distances, and also represented in stellar populations of all ages.

- ☆ Back in Virginia, these various activities by Skrutskie, Wilson, and Majewski evolved the original multifiber NIR spectroscopy concept for SDSS-III. The program became a comprehensive and high spectral resolution survey of Galactic chemodynamics, targeting primarily (though not exclusively) giant stars spanning from the Galactic bulge, across the dust-obscured disk and out into the halo. The 2MASS database, whose creation was led by Skrutskie, was the perfect source catalog for such a survey. Another key aspect of the proposed survey, largely unique to Milky Way surveys, was the inclusion of multi-epoch observing. This strategy served two purposes: (i) to build high S/N spectra of fainter sources and (ii) to identify Doppler-variable stars for the identification of binary and even lower mass stellar companions.
- ☆ APOGEE (/surveys/apogee) was proposed as a project for SDSS-III in 2006 August and was officially approved by the ARC Board as one of the four SDSS-III projects in 2006 November. Selection of the APOGEE wavelength range and resolution was influenced by the available chemical species from known line transitions, the pioneering efforts by Verne Smith (NOIR Lab; e.g., Smith & Lambert 1985 (<https://ui.adsabs.harvard.edu/abs/1985ApJ...294..326S/abstract>)) and later work by Smith et al. 2002 (<https://ui.adsabs.harvard.edu/abs/2002AJ....124.3241S/abstract>) and Cunha & Smith 2006 (<https://ui.adsabs.harvard.edu/abs/2006ApJ...651..491C/abstract>), as well as extensive spectral modeling by Carlos Allende-Prieto (IAC). Because a number of the lines in the selected APOGEE region (1.5-1.7 microns) remained unidentified, the APOGEE team initiated collaborations with laboratory atomic physicists from University of Wisconsin and Imperial College London to resolve this problem.
- ☆ The science goals for APOGEE required the creation of a unique, state-of-the-art spectrograph to undertake highly multiplexed fiber spectroscopy in the H-band. Four new technologies were innovated to make the first APOGEE spectrograph, for which the optical bench is held in a cryogenic and vacuum cryostat to approximately liquid nitrogen temperature. The first spectrograph was built on time and budget. Commissioning occurred at APO in Spring 2011. In 2017, APOGEE Instrument Scientist John Wilson and the APOGEE Team were given the Maria and Eric Muhlmann Award (<https://astrosociety.org/who-we-are/awards/maria-and-eric-muhlmann-award.html>) from the Astronomical Society of the Pacific (<https://astrosociety.org/>) for the innovative and successful design of the APOGEE spectrographs.
- ☆ Another innovation was the (*necessary*) automation of the processing and interpretation of the high volume of high-resolution spectra generated by the APOGEE survey. The core of the ASPCAP pipeline is Carlos Allende Prieto's optimization code, FERRE. Led by Jon Holtzman and David Nidever, a new code was developed to derive stellar parameters and stellar abundances efficiently and automatically using FERRE.
- ☆ Official APOGEE-1 survey data acquisition commenced in September 2011. It concluded three years later in July 2014, exceeding its original goal to survey 100,000 stars to H=12.2 mag with S/N>100. The first release of APOGEE-1 data occurred in DR10, followed by two subsequent releases, DR12 (July 2015) and DR13 (July 2016).
- ☆ As part of SDSS-IV, it was proposed to expand APOGEE-2 to the Southern Hemisphere, which provided a much better view of the Galactic bulge and inner disk and enabled the expansion of APOGEE to the Magellanic Clouds. A second spectrograph was built for the du Pont telescope, which was delivered to Las Campanas Observatory.
- ☆ APOGEE-2 started Northern Hemisphere operations immediately after APOGEE-1 finished in 2014 and concluded in November 2020. The Southern Hemisphere operations for APOGEE-2 began in early 2017 and concluded in January 2021. An additional change is that APOGEE-2N not only observes in bright time but co-observes with the SDSS-IV MaNGA survey (/surveys/manga/) during the dark time.
- ☆ After the first few years of very efficient operations and fortuitously good weather, APOGEE-2 North observations were already a year ahead of schedule. The efficiency prompted the creation of the “Bright Time Extension (BTX)” program, to further expand the breadth of APOGEE-2 observations, spreading its Milky Way coverage as well as including a special focus on the TESS Continuous Viewing Zones, *Kepler* and K2 fields.
- ☆ The first release of APOGEE-2 data took place in DR14 (/dr14/irspec). DR16 (/dr16/irspec) was the first data release to include data from APOGEE-2S. DR17 (/dr17/irspec) is the final release from APOGEE-2 and includes the full observations taken from APOGEE-2S and APOGEE-2N.

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